

SEYYED AMIRHOSSEIN (ARSHIYA) MOSTAFAVISABET

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EDUCATION

University of Toronto

Sep. 2023 – Apr. 2029 (expected)

Bachelor of Applied Science & Engineering in Computer Engineering + PEY Co-op

- **Intended Minors:** Engineering Business
 - **Relevant Courses:** Programming Fundamentals, Digital Logic, Computer Organization, Circuit Analysis, Introductory Electronics, Signals & Systems, Software Design & Communication
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EXPERIENCE

STEMMINDS Corp. | Aurora, ON

Software Engineer

Sep. 2026 – present

R&D Team Member

May. 2026 – Sep. 2026

Technology & Innovation Specialist

May. 2025 – May. 2026

- Developed a modular C++ API for Raspberry Pi-based sensor nodes, supporting 5+ sensor types and reducing the sensor integration setup time by ~80%.
 - Designed and fabricated a custom printed circuit board (PCB) sensor node supporting 6 analog, 4 digital, and 2 I²C sensors using KiCAD and Voltera V1 PCB printer machine, doubling the sensor capacity from 6 to 12.
 - Prototyped and deployed an ultrasonic water tank gauge using an Arduino Leonardo and HM-10 Bluetooth low energy (BLE) module, developing firmware for water percentage measurement, wireless transmission, and interrupt driven low power operation with ~90% inactive time.
 - Deployed a 5-node LoRaWAN network of distributed agricultural sensor nodes using The Things Network server, collecting soil and environmental data for monitoring crop fields and green houses.
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PROJECTS

Haptic Sleeve

Aug. 2026 – present

- Designed and simulated a multi-channel haptic control circuit using Vibration Motors and PCA9685 PWM Servo Driver, implementing current protection and independent control of 16+ motors.
 - Developing a Python Simulation Program to synthesize 5+ physical sensations through signal analysis and encode the resulting signals as PWM inputs for 16+ vibrations motors.
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SKILLS

Programming: C/C++, RISC-V Assembly, Arduino

Hardware & Embedded: Circuit Design, Digital Logic, Microcontrollers, FPGA/SoC, PCB Design, Hardware Troubleshooting

Design & Simulation: KiCAD, LTSpice

Fabrication: Voltera V1 PCB Printer

Professional: Technical Documentation, Team Collaboration, Project Management, Problem Solving